

EcoTrack Lite

Energy Awareness for Shared Student Living

Promoting Sustainable Energy Habits Among Students in Shared Apartments

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Why this matters

- Energy waste increases electricity costs.
- Students in shared apartments split bills equally.
- Actual energy usage is not visible.
- Wasteful behavior often goes unnoticed.

Target Group

- Students living in shared apartments (WG)
- International students in Germany
- Apartments without smart meters

Scenario

- Lights and heaters are left on unnecessarily.
- Shared electricity bills increase for everyone.

Problems

- No transparency in energy usage
- Unfair cost sharing
- Difficult to identify waste behavior

Existing Solutions

- JouleBug
- General energy tracking apps

Gap

- No simple solution for shared student living without smart devices

Our Solution: EcoTrack Lite

- Manual energy tracking
- Waste detection
- Personalized tips
- Daily and weekly summaries

👉 Simple, low-cost, and easy to use

How the System Works

Input

- Manual appliance usage data
- Predefined energy values

Detection Logic (Rule-Based)

- Usage exceeds threshold → Waste detected
- Night-time usage → Alert
- High total usage → Warning

Output

- Alerts and recommendations
- Daily and weekly summaries

👉 **Core Technology:** Rule-Based Energy Analysis System

Manual Input for Energy Use (Approx.)

	Device	Power (W)	Daily Use Example	kWh/day
	Laptop	30–100 W	6 hours	0.2–0.6
	Stove	1000–3000 W	1 hour	1–3
	Heater	1000–2500 W	5 hours	5–12.5
	Lights	5–15 W each	5 hours	0.1–0.2
	TV	50–150 W	4 hours	0.2–0.6



Users manually enter daily usage of devices using a predefined appliance list with estimated energy values.

Sustainability Impact



Electricity Consumption for a Student



Key Insight

- For a student in Germany:
- Normal (no heating): cheap, **~80–150 kWh**
- Winter with electric heating: very high, **200–600 kWh**
- Stove + electronics are minor compared to heating



How It Will Work

A student can set a benchmark for how much electricity he needs to consume. If he overuses it, he will receive a notification.



Normal Student Baseline (Example)

- **120 kWh/month**
- If usage > **150 kWh** → Warning
- If usage > **250 kWh** → High Alert



Student Electricity Usage in Germany



Normal Usage
(No heating)

~80–150 kWh
per month



Warning
High Usage

>150 kWh
per month



High Alert
Very High Usage

>250 kWh
per month

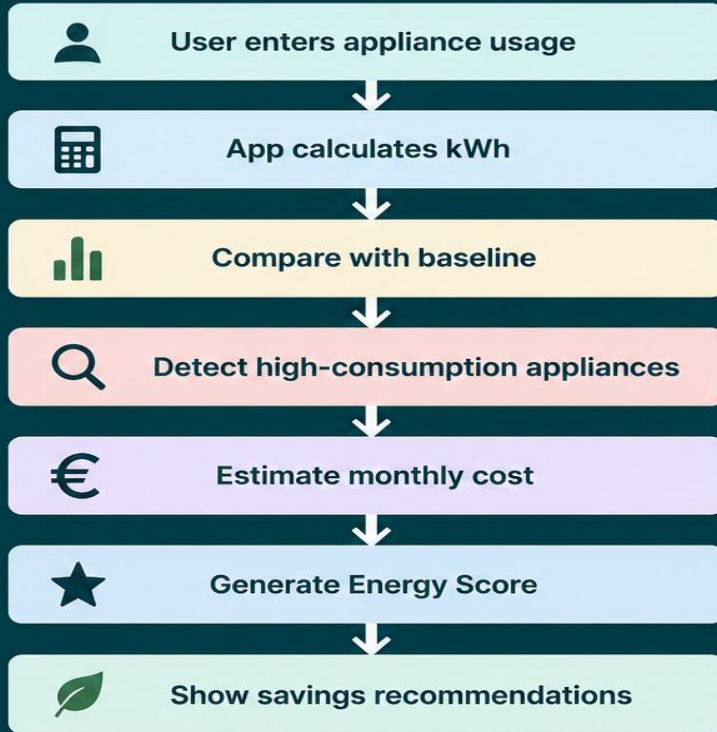


EcoTrack Lite helps students stay within a sustainable energy baseline and reduces unnecessary electricity waste.

Evaluation Strategy and Feasibility



Evaluation Strategy



Evaluation Metrics

- ✓ Reduction in estimated energy consumption
- ✓ Reduction in monthly electricity cost
- ✓ Improvement in Energy Score
- ✓ Number of waste alerts generated
- ✓ User satisfaction and usability feedback



Feasibility

Achievable Aspects

- ✓ Simple web or mobile prototype
- ✓ Manual data input
- ✓ Rule-based energy analysis
- ✓ Achievable within one semester

Not Included

- ✗ No smart meter integration
- ✗ No complex AI required
- ✗ No real-time sensor needed



Focus: Realistic and practical solution for students living in shared apartments in Germany.

Thank You

Any Questions?